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Introduction: Despite careful health screening by NASA, astronauts adapted to microgravity may require the administration of anesthesia for a variety of surgical conditions not only in the highly autonomous environments of space habitation or Mars-type missions but also for immediate surgical care after evacuation from low orbital missions. Physiologic adaptation to weightlessness (e.g., changes in blood and plasma volume, fraction of cardiac output directed to brain, autonomic nervous system activity) can be expected to markedly alter the in vivo behavior and characteristics of anesthetics/hypnotics. A suitable anesthetic technique for microgravity has not been devised. Furthermore, independent external review of the experience with primates on BION 11 has led to the conclusion that "...there was an unexpected mortality risk associated with anesthesia ...following return from space ..." Total intravenous anesthesia (TIVA) is probably the most practical and cost-effective method of rendering subjects insensate during surgery under microgravity conditions. Equipment for TIVA is light, compact, and self-contained. Among the TIVA agents in use today, propofol (2, 6-diisopropylphenol) is the best single agent suitable for use in space. Propofol has several key properties that make it a suitable anesthetic in a space environment: 1) rapid induction of and emergence from anesthesia, 2) ability to adjust the dose to achieve anesthetic states ranging from conscious sedation to general anesthesia, 3) minimal drug accumulation in the body, even during long procedures, 4) significant anti-emetic properties, and 5) a simple and reliable means for assessing the depth of anesthesia/hypnosis, the bispectral index (BIS), derived from the electrical activity of the brain.
Hypotheses: 1. Adaptation to microgravity increases the hypnotic/anesthetic effect of propofol by changing its pharmacokinetics (PK), but not pharmacodynamics (PD). 2. Adaptation to microgravity does not change the recovery of cognitive function after propofol anesthesia.

Specific Aims: • Determine the changes in clearance, volume of distribution, mean residence time and half-life and pattern of metabolites of propofol caused by adaptation to microgravity. • Relate propofol plasma concentrations to clinical and electrophysiological measures of anesthetic drug effect during normal gravity and adaptation to microgravity • Assemble the pharmacokinetic and pharmacodynamic information to provide guidelines for dosing requirements for sedation and anesthesia in microgravity • Determine changes caused by adaptation to microgravity in the time course of the recovery of cognitive function after propofol anesthesia • Assess whether the adaptation to microgravity causes delayed postoperative cognitive dysfunction.

These hypotheses and aims focus on questions of autonomous medical care, specifically risks number 18, 19 and to a lesser extent 22 as they relate to the capabilities/knowledge background required to provide anesthesia care specifically and pharmacotherapy in general.

Research Plan: The evaluation of the effects of simulated microgravity on the anesthetic properties of propofol was conducted sequentially in two groups of subjects. The first group was exposed to a dose escalation protocol to characterize the changes in pharmacodynamics caused by adaptation to microgravity and to identify any prominent, unexpected and novel safety concerns. The second group served to characterize the pharmacokinetics and pharmacodynamics as they apply to a clinical anesthetic as well as the neuro-cognitive recovery after an anesthetic titrated to comparable anesthetic depth in subjects adapted to either normal or microgravity. The dosing of the anesthetic in the second group was guided by the information from the first group and aimed to provide a clinically relevant scenario. Both groups used a randomized cross-over design. That is, each subject received the same dosing protocol of propofol twice in random order, either adapted to normal gravity or after 48 hours of antiorthostatic bed rest to simulate adaptation to microgravity. The hemodynamic effects of the antiorthostatic bed rest were studied by obtaining complete blood counts and transthoracic echocardiograms before and after the 48 hour bed rest. A third group of 10 subjects controlled for changes in the neuro-cognitive test scores that may occur by repeat administration or learning of tested tasks. This group was matched to the volunteers in health status and test intervals but neither underwent neither anesthesia nor adaptation to simulated microgravity.

Results

Twenty volunteers were enrolled to receive propofol anesthesia, ten volunteers for each of the two groups. Of the 20 volunteers in the propofol groups 18 completed the study without complications. Two anesthetics in Group II were stopped, one because of persistent coughing – likely a residual from a respiratory tract infection - and one because of regurgitation without aspiration – most likely caused by inadequate anesthesia. Independent review of all anesthetics by a panel of 4 anesthesiologists as part of Data Safety Monitoring oversight did agree with the investigators that no unexpected novel safety concerns were raised by the study anesthetics. Antiorthostatic bed rest led to hemoconcentration and typical mild symptoms of back pain or a sensation of fullness of the head/ headache. Figure 2 illustrates the magnitude of the changes in the hemogram caused by 48 hours of anti-orthostatic bedrest.

For both propofol groups the anesthetic effect of propofol was similar in normal gravity and simulated microgravity. The clinical assessment by the modified observer's assessment of alertness/sedation scale was corroborated by BIS monitoring. As intended, subjects in group II underwent clinically equivalent anesthetics in normo- and simulated microgravity. Consistent with the expected hemoconcentration, but unexpected from the clinical course, these anesthetics resulted in propofol plasma concentrations for up to 60 minutes after the anesthetic that were higher in simulated microgravity than in normogravity ($p<0.05$, repeated measures ANOVA). While propofol impaired aspects of cognitive function early after the anesthetic, the recovery was the same for normal and simulated microgravity (Figure 8). There was no evidence of delayed postoperative cognitive dysfunction.

Our results suggest only a minor impact of the adaptation to weightlessness on the pharmacodynamic/pharmacokinetic profile of propofol. A second important finding of our study was the absence of novel safety concerns caused by adaptation to microgravity, especially since both primates on BION 11 experienced complications that were thought to be caused by anesthesia. There are important limitations to our study. First, adaptation to microgravity was only brief, i.e. simulated by 48 hours of antiorthostatic bed rest. While this duration is sufficient to cause the changes in blood volume and blood flow characteristic of microgravity, and thereby simulates microgravity-induced changes in distribution and elimination of drugs, it falls short of reproducing the clinical scenario of anesthesia care after a medical evacuation from ISS or autonomous care on a long-term mission. Second, anesthesia was studied in the absence of surgery and surgical disease. Both alter the neuro-humeral milieu and may profoundly influence the patient's response to an anesthetic.

Specific CPR questions addressed by this project are:

18 a What are the essential technologies, resources, procedures, skills, and training necessary to provide effective prevention strategies to mitigate each of the conditions listed in the SMCCB-approved Space Medicine Condition List (catalogued in the online Patient Condition Database)? [ISS 3, Lunar 1, Mars 1] Our research would suggest that initial

	preparations for administration of anesthesia can be patterned after training, procedures, requirements and technologies used on earth. These same preparations should extend to anesthesia-related aspects of the following questions 18d What resources are required for telemedical consultation, diagnosis, and management of major trauma? [ISS 3, Lunar 2, Mars 1] 18ah What resources and procedures are needed for the surgical management of major illness, injury, and trauma? [ISS 3, Lunar 1, Mars 1] 18ak What are effective regional and local anesthesia strategies in reduced G? [ISS TBD, Lunar TBD, Mars TBD] 23a What are the necessary clinical skills/knowledge for a space medicine physician? [ISS 4, Lunar 1, Mars 1] 22a What decision support technologies are needed to support clinical care? [ISS 4, Lunar 2, Mars 1] On a separate level, our study addresses aspects of the PK/PD changes caused by adaptation to microgravity as stated in question by studying a drug with well defined rapid effects in a strict crossover design. This question is raised in the CPR as question: 19a What are the effects of space flight and reduced-G on the absorption, distribution, metabolism, clearance, excretion, clinical efficacy, side effects and drug interactions for medications used in primary, secondary and tertiary prevention of conditions stated in the Space Medicine Condition List? [ISS 2, Lunar 2, Mars 1] Our results suggest small changes in PK/PD that fall within the spectrum of normal inter-individual variability are caused by adaptation to microgravity at least for drugs that are administered intravenously and relatively lipophilic. The Countermeasure Readiness level is 4-7.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study proactively addressed a question relevant to space flight in an area where previous experience (BION 11) suggested serious problems. There are no direct earth benefits.
Task Progress:	Our study is essentially complete as it was originally proposed. Detailed pharmacokinetic analysis is still ongoing but will be completed by August 2007. Twenty volunteers were enrolled to receive propofol anesthesia, ten volunteers for each of the two groups. Of the 20 volunteers in the propofol groups 18 completed the study without complications. Two anesthetics in Group II were stopped, one because of persistent coughing – likely a residual from a respiratory tract infection - and one because of regurgitation without aspiration – most likely caused by inadequate anesthesia. Independent review of all anesthetics by a panel of 4 anesthesiologists as part of Data Safety Monitoring oversight did agree with the investigators that no unexpected novel safety concerns were raised by the study anesthetics. Antiorthostatic bed rest led to hemoconcentration and typical mild symptoms of back pain or a sensation of fullness of the head/ headache. Figure 2 illustrates the magnitude of the changes in the hemogram caused by 48 hours of anti-orthostatic bedrest. For both propofol groups the anesthetic effect of propofol was similar in normal gravity and simulated microgravity. The clinical assessment by the modified observer's assessment of alertness/sedation scale was corroborated by BIS monitoring. As intended, subjects in group II underwent clinically equivalent anesthetics in normo- and simulated microgravity. Consistent with the expected hemoconcentration, but unexpected from the clinical course, these anesthetics resulted in propofol plasma concentrations for up to 60 minutes after the anesthetic that were higher in simulated microgravity than in normogravity ($p<0.05$, repeated measures ANOVA). While propofol impaired aspects of cognitive function early after the anesthetic, the recovery was the same for normal and simulated microgravity (Figure 8). There was no evidence of delayed postoperative cognitive dysfunction. Our results suggest only a minor impact of the adaptation to weightlessness on the pharmacodynamic/pharmacokinetic profile of propofol. A second important finding of our study was the absence of novel safety concerns caused by adaptation to microgravity, especially since both primates on BION 11 experienced complications that were thought to be caused by anesthesia. There are important limitations to our study. First, adaptation to microgravity was only brief, i.e. simulated by 48 hours of antiorthostatic bed rest. While this duration is sufficient to cause the changes in blood volume and blood flow characteristic of microgravity, and thereby simulates microgravity-induced changes in distribution and elimination of drugs, it falls short of reproducing the clinical scenario of anesthesia care after a medical evacuation from ISS or autonomous care on a long-term mission. Second, anesthesia was studied in the absence of surgery and surgical disease. Both alter the neuro-humeral milieu and may profoundly influence the patient's response to an anesthetic.
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